

PAPER ID	:	1571313621
PAPER TITLE	:	<i>A Comprehensive Parasitic-Aware Post-Layout Design and Verification Methodology for a 6.5-GHz Ultra-Low-Power Ring Oscillator in 65-nm CMOS</i>

Reviewer Feedback

Go through each comment one by one — a page or paragraph reference makes it easy for everyone to follow.

NO.	REVIEWER'S FEEDBACK	AUTHOR'S RESPONSE	LOCATION OF REVISION IN MANUSCRIPT
Reviewer 1			
1	The paper is well structured and organized. Moreover, the design process is well explained. Nevertheless, more improvement may be employed in the layout to reduce process mismatch should this device be manufactured in the future for further testing/validation.	Addressed. New Section III-C, "Matching and Mismatch Considerations," states the matching constraints applied and reports the result: the internal nodes net3 and net5 differ by only 0.02 fF out of ~3.5 fF (below 1%). Dummy poly, guard rings, and matched dummy loading are identified as pre-tape-out measures. A full layout revision for random mismatch is stated as future work.	Section Physical Design and Custom Layout Methodology, Sub-section Matching and Mismatch Considerations, Page 3.
Reviewer 2			
1	The design of a ring oscillator to get ultra-low-power consumption is not new. The author should explain in detail why you chose that topology.	Addressed. New Section II-A, "Topology Selection," explains why choose that topology	Page 2, section II-A
2	Parasitic extraction in RFIC design cannot be stated as a proposed framework, since it is a well-established method in chip design. If the authors intend to present it as a proposed framework, they should elaborate on the novelty of their framework.	The claim has been rescoped throughout; the abstract, Introduction, Section V-D, and Conclusion no longer describe the work as a proposed framework. Section V-D now states that the contribution lies neither in a new topology nor in the extraction procedure itself,	Page 1, 2, and 6



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		but in the magnitude of the correction it produces: a schematic-only report would have overstated the frequency by 4.65 GHz.	
3	Please create the schematic figures using other software rather than using screen captures from Cadence.	Fig. 1 has been redrawn using external drawing software (https://app.smartdraw.com)	Page 3
4	For Fig. 4, compare the transient responses of the schematic and post-layout simulations to clearly show the differences.	Not completed within the revision period. The quantitative comparison is already given in Table III (frequency, loop period, stage delay for both cases) and analysed in Section V-B.	-
5	Did the authors perform any layout optimization on the initial layout? If so, please describe the optimization steps that were carried out. To optimize the layout, I suggest first evaluating the transistor core and comparing its performance with that of the schematic. After that, the interconnects can be analyzed and optimized.	A systematic two-step optimisation (device-core extraction, then interconnect analysis) was not carried out for this version. We have adopted the suggested procedure for the next design iteration, now stated in the Conclusion.	Page 6
6	In Table IV, I also recommend that the authors include the pre-layout simulation results for comparison.	Not completed within the revision period, as it requires a separate schematic-level PSS/pnoise run. Table III already gives the pre- versus post-layout comparison, and Section V-D quantifies the 4.65 GHz overstatement a schematic-only report would produce.	-
Reviewer 3			
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Desk Reviewer Feedback

Any editorial notes from the desk review team.

NO.	REVIEWER'S FEEDBACK	AUTHOR'S RESPONSE	LOCATION OF REVISION IN MANUSCRIPT
1	Use keywords instead of index terms. Do not use capital letters for each keyword	Has been fixed by replacing "index terms" with "keywords"	Page 1
2	Please ensure that all citations are presented in the correct ascending numerical order throughout the manuscript	It's confirmed that all citations are presented in correct ascending numerical order	Page 1
3	Some figures and table had not been cited in the manuscript. Directly cite all figures and tables within the text of your paper. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 1", even at the beginning of a sentence	all tables and figures have been cited in manuscript	Page 3-6
4	Place figures and tables at the top and/or bottom of columns. Avoid placing them in the middle of columns	Have been fixed	Page 3-6
5	In references section, please format the reference following the IEEE style. Especially for jurnal and proceeding reference template	It is ensured that the reference format following the IEEE style	At references section, page 6
6	Please carefully review and revise the manuscript to ensure full compliance with the IEEE paper formatting guidelines. The official IEEE templates and instructions can be accessed at: https://www.ieee.org/conferences/publishing/templates	Have been fixed	

OVERALL COMMENTS (OPTIONAL)

